
Lab 3

Task 1 - CNCF distribution:

```
[root@serverb ~]# docker run -d -p 5000:5000 --name Local-registry registry:2
Unable to find image 'registry:2' locally
2: Pulling from library/registry
6d464ea18732: Pull complete
44cf07d57ee4: Pull complete
bbbdd6c6894b: Pull complete
3493bf46cdec: Pull complete
8e82f80af0de: Pull complete
b537bf6d1146: Download complete
32a76c78501f: Download complete
Digest: sha256:a3d8aaa63ed8681a604f1dea0aa03f100d5895b6a58ace528858a7b332415373
Status: Downloaded newer image for registry:2
59e53b97231b952adbc17d7eb5920d1cc72b9946d8dd1201213eab5dae872e36
[root@serverb ~]#
```

Running the registry container on port 5000.

```
[root@serverb nginx-image]# cat Dockerfile
FROM alpine:latest

RUN apk add --no-cache nginx && mkdir -p /run/nginx

ENTRYPOINT ["nginx","-g","daemon off;"]
[root@serverb nginx-image]#
```

The Dockerfile is created to install the nginx and create the /run/nginx directory which nginx will use to store its pid file.

The nginx -g daemon off; command will set the nginx process to work in the foreground to keep the container running.

```

[root@serverb nginx-image]# docker build -t localhost:5000/nginx-alpine .
[+] Building 7.7s (6/6) FINISHED                                docker:default
=> [internal] load build definition from Dockerfile             0.6s
=> => transferring dockerfile: 210B                             0.1s
=> [internal] load metadata for docker.io/library/alpine:latest 0.4s
=> [internal] load .dockerignore                               0.0s
=> => transferring context: 2B                                    0.0s
=> CACHED [1/2] FROM docker.io/library/alpine:latest@sha256:25109184c71b 0.2s
=> => resolve docker.io/library/alpine:latest@sha256:25109184c71bdad752c 0.1s
=> [2/2] RUN apk add --no-cache nginx && mkdir -p /run/nginx    5.1s
=> exporting to image                                           0.7s
=> => exporting layers                                           0.5s
=> => exporting manifest sha256:24b6d6320468a8bb272ebb023c9dfc4d02dc23a3 0.0s
=> => exporting config sha256:5ff0e19ff2334445cdb0a8f352f3b83329df520e93 0.0s
=> => exporting attestation manifest sha256:b9b3ff49fb39c9868caadf3dc18a 0.0s
=> => exporting manifest list sha256:2973f2206a470031aec9d4040613d6455ae 0.0s
=> => naming to localhost:5000/nginx-alpine:latest             0.0s
=> => unpacking to localhost:5000/nginx-alpine:latest          0.1s

```

Building the image with the name localhost:5000/nginx-alpine.

```

[root@serverb nginx-image]# docker push localhost:5000/nginx-alpine:latest
The push refers to repository [localhost:5000/nginx-alpine]
589002ba0eae: Pushed
37fe23751094: Pushed
04d67e6c8487: Pushed
latest: digest: sha256:2973f2206a470031aec9d4040613d6455ae404f5b7711142a5c1f97e8c758b7d size: 855
[root@serverb nginx-image]#

```

The image is pushed to the local registry.

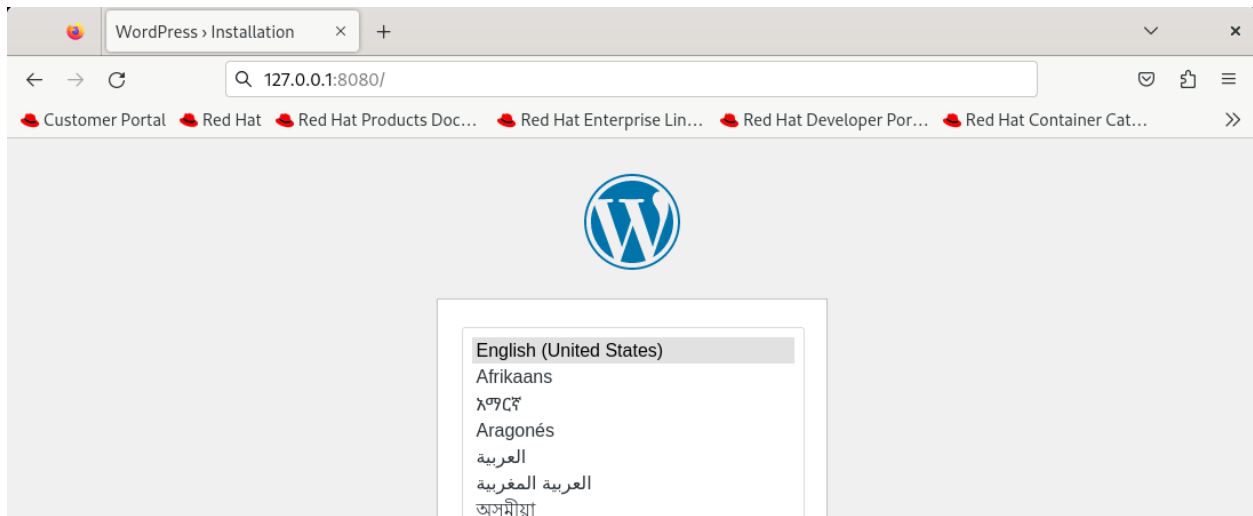
Task 2 – WordPress application:

```
[root@serverb wordpress-app]# cat docker-compose.yml
services:
  DB-mysql:
    image: mysql:5.7
    environment:
      MYSQL_DATABASE: wordpressDB
      MYSQL_USER: wp_user
      MYSQL_PASSWORD: wp_password
      MYSQL_ROOT_PASSWORD: rootpass
    volumes:
      - /var/lib/mysql:/var/lib/mysql

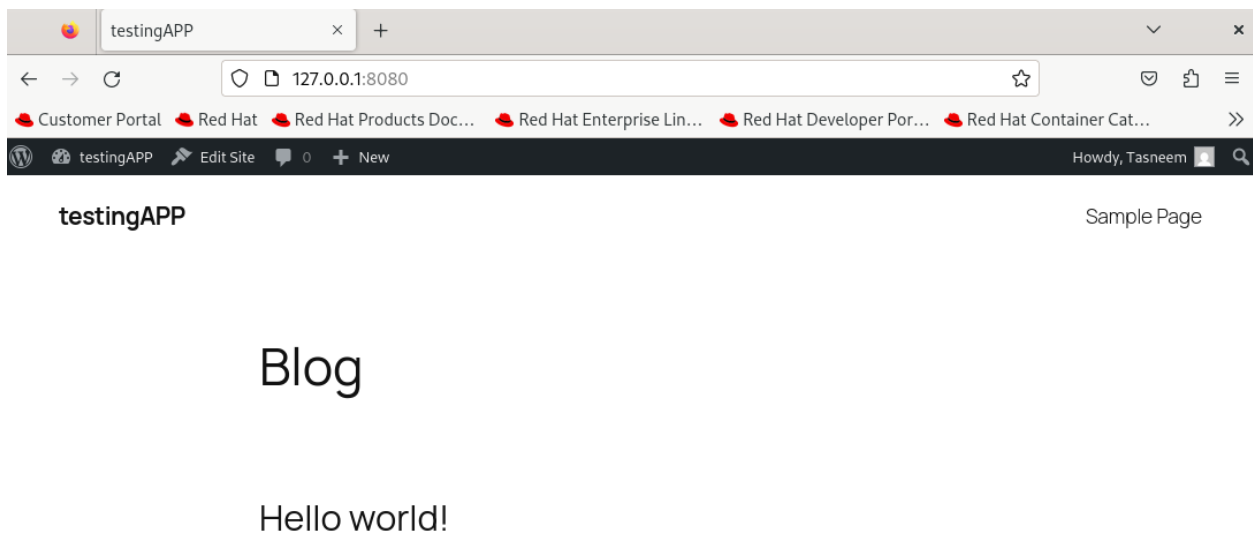
  wordpressAPP:
    depends_on:
      - DB-mysql
    image: wordpress:latest
    ports:
      - "8080:80"
    environment:
      WORDPRESS_DB_HOST: DB-mysql:3306
      WORDPRESS_DB_USER: wp_user
      WORDPRESS_DB_PASSWORD: wp_password
      WORDPRESS_DB_NAME: wordpressDB
```

The docker-compose file with the 2 services (containers).

```
[root@serverb wordpress-app]# docker compose up
[+] up 43/43
✓ Image mysql:5.7 Pulled 205.5s
✓ Image wordpress:latest Pulled 217.1s
✓ Network wordpress-app_default Created 4.7s
✓ Container wordpress-app-DB-mysql-1 Created 2.7s
✓ Container wordpress-app-wordpressAPP-1 Created 0.5s
Attaching to DB-mysql-1, wordpressAPP-1
DB-mysql-1 | 2026-03-03 08:48:43+00:00 [Note] [Entrypoint]: Entrypoint script for MySQL Server 5.7.44-1.el7 started.
DB-mysql-1 | 2026-03-03 08:48:44+00:00 [Note] [Entrypoint]: Switching to dedicated user 'mysql'
DB-mysql-1 | 2026-03-03 08:48:45+00:00 [Note] [Entrypoint]: Entrypoint script for MySQL Server 5.7.44-1.el7 started.
wordpressAPP-1 | WordPress not found in /var/www/html - copying now...
DB-mysql-1 | 2026-03-03 08:48:45+00:00 [Note] [Entrypoint]: Initializing database files
DB-mysql-1 | 2026-03-03T08:48:45.936777Z 0 [Warning] TIMESTAMP with implicit DEFAULT value is deprecated. Please use --explicit_defaults_for_timestamp server option (see documentation for more details).
DB-mysql-1 | 2026-03-03T08:48:46.912636Z 0 [Warning] InnoDB: New log files created, LSN=45790
DB-mysql-1 | 2026-03-03T08:48:47.039366Z 0 [Warning] InnoDB: Creating foreign key constraint system tables.
DB-mysql-1 | 2026-03-03T08:48:47.189566Z 0 [Warning] No existing UUID has been found, so we assume that this is the first time
that this server has been started. Generating a new UUID: cad76c3a-16dd-11f1-9251-420bb5dcf9fe.
DB-mysql-1 | 2026-03-03T08:48:47.199620Z 0 [Warning] Gtid table is not ready to be used. Table 'mysql.gtid_executed' cannot be
opened.
DB-mysql-1 | 2026-03-03T08:48:48.308504Z 0 [Warning] A deprecated TLS version TLSv1 is enabled. Please use TLSv1.2 or higher.
```



The WordPress is working.



A testingAPP web page is created.

```
[root@serverb wordpress-app]# ls /var/lib/mysql/wordpressDB/
db.opt          wp_links.frm      wp_postmeta.ibd   wp_term_relationships.frm  wp_term_taxonomy.ibd
wp_commentmeta.frm wp_links.ibd      wp_posts.frm      wp_term_relationships.ibd  wp_usermeta.frm
wp_commentmeta.ibd wp_options.frm    wp_posts.ibd      wp_terms.frm              wp_usermeta.ibd
wp_comments.frm   wp_options.ibd    wp_termmeta.frm   wp_terms.ibd              wp_users.frm
wp_comments.ibd   wp_postmeta.frm   wp_termmeta.ibd   wp_term_taxonomy.frm      wp_users.ibd
[root@serverb wordpress-app]#
```

The /var/lib/mysql is mounted successfully from the host, and it now has the WordPress application data.

Task 3 – Bonus:

```
[root@serverb wordpress-app]# docker image tag tasneemadel/basic-flask-app:latest localhost:5000/basic-flask-app
[root@serverb wordpress-app]# docker push localhost:5000/basic-flask-app:latest
The push refers to repository [localhost:5000/basic-flask-app]
e19959d4eecb: Pushed
e80c6a38a12e: Pushed
589002ba0eae: Mounted from nginx-alpine
1fce7f734d8e: Pushed
75a963d33c55: Pushed
08634a37813b: Pushed
latest: digest: sha256:b47c6bc80d0de4e22cff6ef3f3afbbaaf812b376b7a251ceb40015d43bbb61da size: 856
[root@serverb wordpress-app]#
```

Tagging the image with the localhost:5000 registry and pushing it to the local registry.

```
[root@serverb wordpress-app]# docker images
```

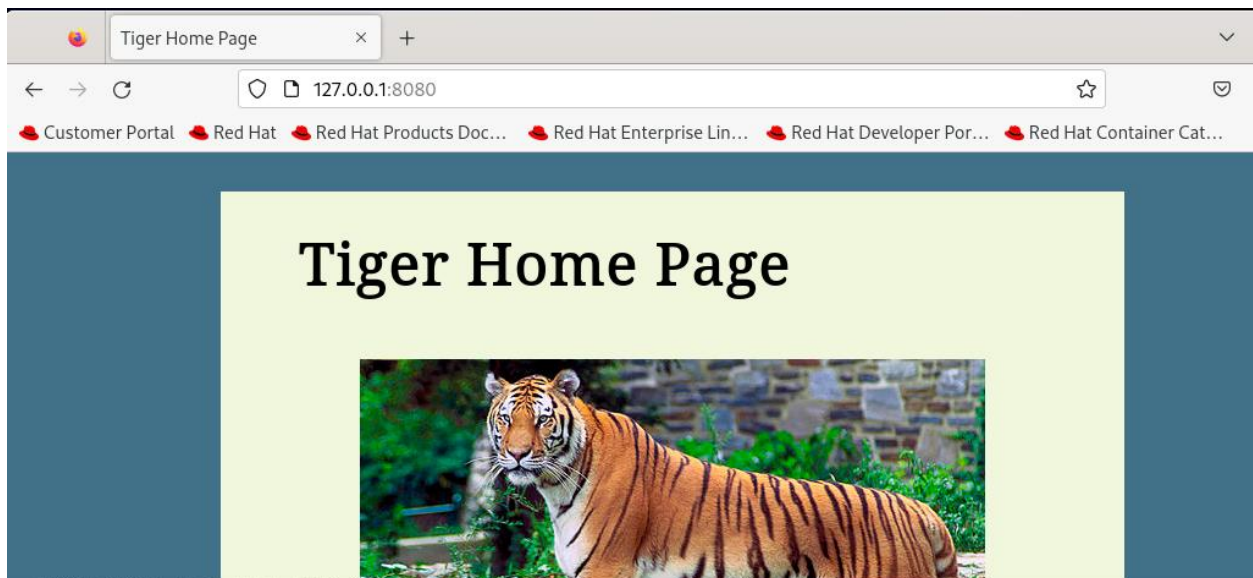
IMAGE	ID	DISK USAGE	CONTENT SIZE	EXTRA
alpine:latest	25109184c71b	12.7MB	3.95MB	
httpd:alpine	8f26f33a7002	95.6MB	22.1MB	
httpd:latest	96b1e8f69ee3	174MB	47.6MB	
localhost:5000/nginx-alpine:latest	2973f2206a47	15.6MB	4.74MB	
mysql:5.7	4bc6bc963e6d	696MB	149MB	
nginx:alpine	1d13701a5f9f	91.9MB	26.9MB	
registry:2	a3d8aaa63ed8	36.5MB	10.3MB	U
webserver-iti-im:latest	8acef0fa3418	172MB	45.2MB	U
wordpress:latest	ffef0dca1f0f	1.07GB	275MB	

```
[root@serverb wordpress-app]#
```

Deleting the flask app image tags from the host to be able to pull it from the local registry.

```
services:
  flask-app:
    image: localhost:5000/basic-flask-app
    ports:
      - "8080:5000"
```

```
[root@serverb basic-flask-app]# docker compose up
[+] up 8/8
✓ Image localhost:5000/basic-flask-app Pulled 13.3s
✓ Network basic-flask-app_default Created 0.8s
✓ Container basic-flask-app-flask-app-1 Created 0.9s
Attaching to flask-app-1
flask-app-1 | * Serving Flask app 'routes'
flask-app-1 | * Debug mode: on
flask-app-1 | WARNING: This is a development server. Do not use it in a production deployment. Use a production WSGI server instead.
flask-app-1 | * Running on all addresses (0.0.0.0)
flask-app-1 | * Running on http://127.0.0.1:5000
flask-app-1 | * Running on http://172.18.0.2:5000
flask-app-1 | Press CTRL+C to quit
flask-app-1 | * Restarting with stat
flask-app-1 | * Debugger is active!
flask-app-1 | * Debugger PIN: 347-911-338
flask-app-1 | 172.18.0.1 - - [03/Mar/2026 09:46:04] "GET / HTTP/1.1" 200 -
flask-app-1 | 172.18.0.1 - - [03/Mar/2026 09:46:04] "GET /static/css/main.css HTTP/1.1" 200 -
flask-app-1 | 172.18.0.1 - - [03/Mar/2026 09:46:04] "GET /static/images/tiger.jpg HTTP/1.1" 200 -
[root@serverb basic-flask-app]#
```



The app is working with the image getting pulled from the local registry.

```
events{}

http{
    server{
        listen 80;
        location / {
            proxy_pass http://flask-app:5000;
            proxy_set_header Host $host;
            proxy_set_header X-Real-IP $remote_addr;
        }
    }
}

~
~
~
~
~
~
~
```

The file nginx.conf is created in the project directory, having the configuration that tells nginx to pass any path to flask-app:5000 container.

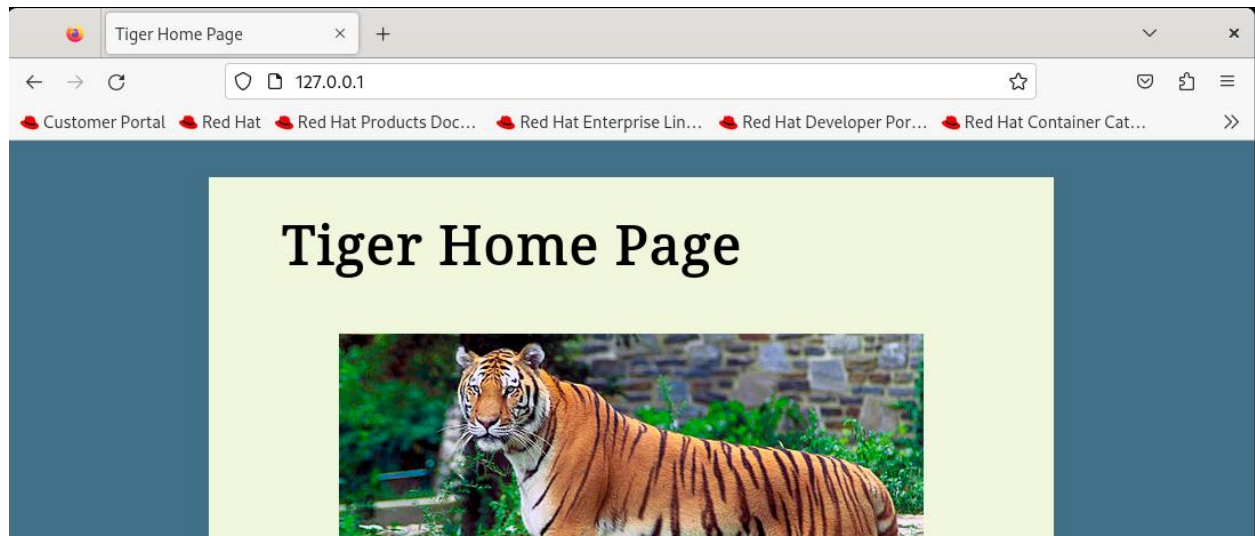
```
root@serverb:~/basic-flask-app
services:
  flask-app:
    image: localhost:5000/basic-flask-app

  nginxServer:
    image: nginx:alpine
    ports:
      - "80:80"
    volumes:
      - ./nginx.conf:/etc/nginx/nginx.conf:ro
    depends_on:
      - flask-app
```

The docker compose file now having the nginxServer container with the nginx.conf file being mounted to the nginx.conf inside the nginx container.

```
[root@serverb basic-flask-app]# docker compose up
[+] up 3/3
✓ Network basic-flask-app_default Created 0.4s
✓ Container basic-flask-app-flask-app-1 Created 0.4s
✓ Container basic-flask-app-nginxServer-1 Created 0.3s
Attaching to flask-app-1, nginxServer-1
nginxServer-1 | /docker-entrypoint.sh: /docker-entrypoint.d/ is not empty, will attempt to perform configuration
nginxServer-1 | /docker-entrypoint.sh: Looking for shell scripts in /docker-entrypoint.d/
nginxServer-1 | /docker-entrypoint.sh: Launching /docker-entrypoint.d/10-listen-on-ipv6-by-default.sh
nginxServer-1 | 10-listen-on-ipv6-by-default.sh: info: Getting the checksum of /etc/nginx/conf.d/default.conf
nginxServer-1 | 10-listen-on-ipv6-by-default.sh: info: Enabled listen on IPv6 in /etc/nginx/conf.d/default.conf
nginxServer-1 | /docker-entrypoint.sh: Sourcing /docker-entrypoint.d/15-local-resolvers.envsh
nginxServer-1 | /docker-entrypoint.sh: Launching /docker-entrypoint.d/20-envsubst-on-templates.sh
nginxServer-1 | /docker-entrypoint.sh: Launching /docker-entrypoint.d/30-tune-worker-processes.sh
nginxServer-1 | /docker-entrypoint.sh: Configuration complete; ready for start up
flask-app-1 | * Serving Flask app 'routes'
flask-app-1 | * Debug mode: on
flask-app-1 | WARNING: This is a development server. Do not use it in a production deployment. Use a production WSGI server instead.
flask-app-1 | * Running on all addresses (0.0.0.0)
flask-app-1 | * Running on http://127.0.0.1:5000
flask-app-1 | * Running on http://172.18.0.2:5000
```

The containers have been created.



The tiger page is loaded successfully.